

Lecture Notes on Syzygies

A Simple but Penetrating Introduction

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1 Why study syzygies?

In algebra, geometry, and topology, we often begin with *generators*. A module, an ideal, or a ring may be described by a list of elements from which everything else is built. But generators are only the first layer of structure. The next question is:

How are the generators related to one another?

A **syzygy** is exactly such a relation.

This idea is fundamental because mathematics often proceeds in layers:

generators $\longrightarrow$ relations $\longrightarrow$ relations among relations $\longrightarrow \dots$

The word *syzygy* refers to one of these layers of relations.

These notes explain the idea carefully, starting with elementary examples and moving toward free resolutions and geometry.

2 First definition

Definition 2.1. Let R be a ring and let $f_1, \dots, f_r \in R$. A **syzygy** among $f_1, \dots, f_r$ is a tuple

$$(a_1, \dots, a_r) \in R^r$$

such that

$$a_1 f_1 + \dots + a_r f_r = 0.$$

So a syzygy is a relation among the elements $f_1, \dots, f_r$, where the coefficients a_i are taken from the ring R .

Remark 2.2. This looks like linear dependence, but over a ring rather than a field. That is why the theory is richer than ordinary linear algebra.

3 The first key example

Let

$$R = k[x, y]$$

for a field k . Consider the two elements

$$f_1 = x, \quad f_2 = y.$$

A syzygy is a pair $(a, b) \in R^2$ such that

$$ax + by = 0.$$

3.1 A visible syzygy

Take

$$a = -y, \quad b = x.$$

Then

$$(-y)x + xy = -xy + xy = 0.$$

So

$$(-y, x)$$

is a syzygy of x and y .

This is the first example one should remember.

3.2 Why it matters

At first, x and y look like two independent generators. But the pair $(-y, x)$ shows that they are tied together by a structured cancellation:

$$(-y) \cdot x + x \cdot y = 0.$$

This is the essence of a syzygy.

3.3 All syzygies of x and y

Now suppose

$$ax + by = 0$$

for some $a, b \in k[x, y]$. We claim that every such relation is a multiple of $(-y, x)$.

Proposition 3.1. *In $R = k[x, y]$,*

$$\text{Syz}(x, y) = R \cdot (-y, x).$$

That is, every syzygy of x and y has the form

$$(-cy, cx)$$

for some $c \in R$.

Proof. Suppose

$$ax + by = 0.$$

Then

$$ax = -by.$$

Hence ax is divisible by y . Since x and y are relatively prime in $k[x, y]$, it follows that a must be divisible by y . So we may write

$$a = -cy$$

for some $c \in R$. Substituting this into the relation gives

$$(-cy)x + by = 0,$$

so

$$y(-cx + b) = 0.$$

Since $k[x, y]$ is an integral domain, we get

$$-cx + b = 0,$$

hence

$$b = cx.$$

Therefore

$$(a, b) = (-cy, cx) = c(-y, x).$$

□

Remark 3.2. This proposition is extremely important conceptually. There may be infinitely many relations, but they are all generated by one basic relation.

4 The module of syzygies

The previous example suggests that syzygies themselves form an algebraic object.

Definition 4.1. Let R be a ring and $f_1, \dots, f_r \in R$. The set of all syzygies among them is called the **syzygy module**:

$$\text{Syz}(f_1, \dots, f_r) := \{(a_1, \dots, a_r) \in R^r \mid a_1 f_1 + \dots + a_r f_r = 0\}.$$

Proposition 4.2. $\text{Syz}(f_1, \dots, f_r)$ is an R -submodule of R^r .

Proof. If

$$(a_1, \dots, a_r), (b_1, \dots, b_r) \in \text{Syz}(f_1, \dots, f_r),$$

then

$$\sum a_i f_i = 0, \quad \sum b_i f_i = 0.$$

Hence

$$\sum (a_i + b_i) f_i = \sum a_i f_i + \sum b_i f_i = 0.$$

So the sum is again a syzygy.

If $c \in R$, then

$$\sum (ca_i) f_i = c \sum a_i f_i = c \cdot 0 = 0.$$

So scalar multiples are again syzygies.

□

5 Syzygies as kernels

The cleanest way to understand syzygies is through kernels.

Given $f_1, \dots, f_r \in R$, define a map

$$\varphi : R^r \rightarrow R, \quad (a_1, \dots, a_r) \mapsto a_1 f_1 + \dots + a_r f_r.$$

Then

$$\text{Syz}(f_1, \dots, f_r) = \ker(\varphi).$$

Thus syzygies are not an isolated notion: they are simply kernels of natural maps.

5.1 The example x, y

For $f_1 = x, f_2 = y$, the map is

$$\varphi : R^2 \rightarrow R, \quad \varphi(a, b) = ax + by.$$

Then

$$\ker(\varphi) = \text{Syz}(x, y) = R \cdot (-y, x).$$

If we think of the ideal $(x, y) \subseteq R$, then φ lands in (x, y) , and we get an exact sequence

$$0 \rightarrow R \xrightarrow{\psi} R^2 \xrightarrow{\varphi} (x, y) \rightarrow 0,$$

where

$$\psi(c) = (-cy, cx).$$

This exact sequence means:

- R^2 records the two generators x, y ,
- the map $R^2 \rightarrow (x, y)$ sends the standard basis vectors to those generators,
- the kernel is exactly the relation among them.

6 A second example: three generators

Now let us look at

$$f_1 = x^2, \quad f_2 = xy, \quad f_3 = y^2$$

in $R = k[x, y]$.

We seek triples $(a, b, c) \in R^3$ such that

$$ax^2 + bxy + cy^2 = 0.$$

Two obvious syzygies are:

$$(-y, x, 0)$$

because

$$(-y)x^2 + x(xy) + 0 \cdot y^2 = -x^2y + x^2y = 0,$$

and

$$(0, -y, x)$$

because

$$0 \cdot x^2 + (-y)(xy) + xy^2 = -xy^2 + xy^2 = 0.$$

So the generators x^2, xy, y^2 satisfy at least these two relations.

Remark 6.1. This example shows that once one has more generators, several basic syzygies may appear.

6.1 A useful perspective

Notice that x^2, xy, y^2 are not random: they are the degree-2 monomials in two variables. Their syzygies come from the fact that the products

$$x \cdot (xy) = y \cdot (x^2), \quad x \cdot (y^2) = y \cdot (xy)$$

coincide. So syzygies often arise from “different ways of writing the same product.”

7 Syzygies versus linear algebra

It is useful to compare syzygies with ordinary linear dependence.

7.1 Linear algebra

If $v_1, \dots, v_r$ are vectors in a vector space over a field k , a linear relation is

$$c_1 v_1 + \dots + c_r v_r = 0, \quad c_i \in k.$$

7.2 Syzygies

If $f_1, \dots, f_r \in R$, then a syzygy is

$$a_1 f_1 + \dots + a_r f_r = 0, \quad a_i \in R.$$

The key difference is that the coefficients a_i are elements of the ring, not scalars from a field. Thus the coefficients themselves may vary polynomially.

Remark 7.1. A syzygy is therefore like a “linear dependence with variable coefficients.”

8 Generators, relations, and presentation

Suppose M is an R -module generated by $m_1, \dots, m_r$. Then there is a surjective map

$$\pi : R^r \rightarrow M, \quad (a_1, \dots, a_r) \mapsto a_1 m_1 + \dots + a_r m_r.$$

Its kernel consists of all relations among the generators $m_1, \dots, m_r$.

Definition 8.1. The kernel $\ker(\pi)$ is called the **module of first syzygies** of the chosen generators.

Thus one has an exact sequence

$$0 \rightarrow \ker(\pi) \rightarrow R^r \rightarrow M \rightarrow 0.$$

This is called a **presentation** of M : one describes M by generators and relations.

8.1 A simple slogan

A syzygy is a relation among generators.

9 Higher syzygies

The first syzygies may themselves have relations.

Suppose

$$0 \rightarrow K_1 \rightarrow F_0 \rightarrow M \rightarrow 0$$

with F_0 free. If K_1 is generated by some elements, then we get a surjection

$$F_1 \rightarrow K_1 \rightarrow 0$$

from another free module F_1 . The kernel of $F_1 \rightarrow K_1$ consists of relations among the first syzygies. These are called **second syzygies**.

Continuing this way, one gets

$$\cdots \rightarrow F_2 \rightarrow F_1 \rightarrow F_0 \rightarrow M \rightarrow 0.$$

Definition 9.1. A sequence

$$\cdots \rightarrow F_2 \rightarrow F_1 \rightarrow F_0 \rightarrow M \rightarrow 0$$

with each F_i free and exact at every stage is called a **free resolution** of M .

Interpretation:

Module	Meaning
F_0	generators of M
F_1	first syzygies
F_2	syzygies among first syzygies
F_3	syzygies among second syzygies
$\vdots$	$\vdots$

10 A concrete free resolution

Let $R = k[x, y]$ and consider the ideal

$$I = (x, y).$$

Then the map

$$\varphi : R^2 \rightarrow I, \quad \varphi(a, b) = ax + by$$

is surjective, and its kernel is generated by $(-y, x)$.

Hence we have an exact sequence

$$0 \rightarrow R \xrightarrow{\psi} R^2 \xrightarrow{\varphi} I \rightarrow 0,$$

where

$$\psi(c) = (-cy, cx).$$

This is a free resolution of I .

It says:

- I has two generators, namely x and y ,
- those two generators satisfy one basic relation,
- there are no further nontrivial relations among that relation in this case.

11 Why syzygies matter in geometry

Syzygies are not merely algebraic bookkeeping. They carry geometric meaning.

11.1 Embedded varieties

Suppose a projective variety $X \subseteq \mathbb{P}^n$ is defined by homogeneous polynomials

$$f_1, \dots, f_r.$$

Then the ideal

$$I(X) = (f_1, \dots, f_r)$$

records the equations cutting out X .

But the equations themselves may satisfy relations:

$$a_1 f_1 + \dots + a_r f_r = 0.$$

These relations are syzygies among the defining equations.

11.2 Geometric meaning

The equations tell us which points lie on X . The syzygies tell us how those equations fit together.

Thus:

$$\textit{Syzygies reveal hidden structure behind the visible equations.}$$

11.3 A guiding picture

$$\text{variety} \longleftrightarrow \text{ideal of equations} \longleftrightarrow \text{syzygies of equations}$$

In modern algebraic geometry, syzygies are used to study:

- embeddings of curves and surfaces,
- complexity of defining equations,
- canonical curves,
- projective normality,
- Koszul cohomology.

12 A simple geometric example

Consider the parabola in $\mathbb{A}^2$:

$$y - x^2 = 0.$$

Its ideal is generated by one equation:

$$I = (y - x^2).$$

Since there is only one generator, there is no nontrivial syzygy among the generators of this ideal.

Now compare with the parametrized curve

$$x = s^2, \quad y = st, \quad z = t^2$$

in projective coordinates. Its image in $\mathbb{P}^2$ is cut out by

$$y^2 - xz = 0.$$

Again the ideal is principal, so there is no first syzygy among generators.

But if an ideal is generated by several polynomials, then their interrelations become important. For more complicated curves, these syzygies encode real geometry.

13 A penetrating interpretation

Let us return to the example

$$x, y \in k[x, y].$$

The syzygy

$$(-y, x)$$

means:

$$(-y) \cdot x + x \cdot y = 0.$$

What does this really say?

It says there are *two ways* to build the same product xy :

$$y \cdot x = x \cdot y.$$

The syzygy records the difference between these two expressions, and that difference is zero.

So a syzygy often measures:

$$\textit{two different constructions that produce the same result.}$$

This viewpoint is extremely useful.

14 Connection with ideals

If $I = (f_1, \dots, f_r) \subseteq R$, then the generators f_i determine a surjection

$$R^r \rightarrow I, \quad e_i \mapsto f_i,$$

where $e_1, \dots, e_r$ is the standard basis of R^r .

Then the syzygy module is

$$\text{Syz}(f_1, \dots, f_r) = \ker(R^r \rightarrow I).$$

Thus the study of ideals naturally leads to the study of syzygies. To know the generators of an ideal is not yet to fully understand the ideal. One also wants to know how the generators interact.

15 Minimal generators and minimal syzygies

In practice one often wants the smallest possible generating set.

Definition 15.1. A generating set $f_1, \dots, f_r$ of an ideal or module is called **minimal** if no generator can be omitted.

Once one has chosen minimal generators, one may seek minimal generators for the syzygy module. These are called **minimal first syzygies**. Then one may continue and seek minimal higher syzygies.

This leads to **minimal free resolutions**, which are among the most refined algebraic invariants of a module.

16 The word “syzygy”

Historically, the word *syzygy* comes from Greek and means something like “yoked together” or “aligned.” In astronomy it refers to celestial bodies lying in a line.

The mathematical usage is apt: generators in a syzygy are tied together by a relation. They are not free-standing; they are aligned by a constraint.

17 A short dictionary

Object	Meaning
Generator	A building block from which the module or ideal is produced
Relation	An expression showing that the generators are not independent
Syzygy	A relation among generators
First syzygy	A relation among the original generators
Second syzygy	A relation among the first syzygies
Free resolution	A layered record of generators, relations, relations among relations, etc.

18 Exercises

Exercise 18.1. Let $R = k[x, y]$. Show directly that

$$(-y, x)$$

is a syzygy of x and y .

Exercise 18.2. Let $R = k[x, y]$. Prove that every syzygy of x and y is a multiple of $(-y, x)$.

Exercise 18.3. Find two explicit syzygies among

$$x^2, \quad xy, \quad y^2.$$

Exercise 18.4. Let $R = k[x, y, z]$. Show that

$$(-y, x, 0), \quad (-z, 0, x), \quad (0, -z, y)$$

are syzygies of x, y, z .

Exercise 18.5. Let $M = R/(x, y)$ with $R = k[x, y]$. Write down a presentation of M and describe the first syzygy module.

Exercise 18.6. Suppose $f_1, \dots, f_r \in R$. Show that $\text{Syz}(f_1, \dots, f_r)$ is an R -submodule of R^r .

19 Solutions sketch

Exercise 1

Compute:

$$(-y)x + xy = -xy + xy = 0.$$

Exercise 2

Assume $ax + by = 0$. Then $ax = -by$, so $y \mid ax$. Since x and y are relatively prime, $y \mid a$. Write $a = -cy$. Then substitute and deduce $b = cx$.

Exercise 3

Two syzygies are:

$$(-y, x, 0), \quad (0, -y, x).$$

Exercise 4

Check directly that each given triple annihilates (x, y, z) under the map

$$R^3 \rightarrow R, \quad (a, b, c) \mapsto ax + by + cz.$$

Exercise 5

The quotient $M = R/(x, y)$ has presentation

$$R^2 \rightarrow R \rightarrow R/(x, y) \rightarrow 0,$$

where $R^2 \rightarrow R$ sends $e_1 \mapsto x$, $e_2 \mapsto y$. The kernel is generated by $(-y, x)$.

20 Final summary

The idea of syzygy is simple but deep.

- Start with generators.
- Ask for relations among them.
- Those relations are syzygies.
- Then ask for relations among those relations.
- This leads to free resolutions.

The basic example

$$(-y)x + xy = 0$$

already contains the whole philosophy: generators may look separate, but in reality they are bound together by hidden structure.

That hidden structure is what syzygies measure.